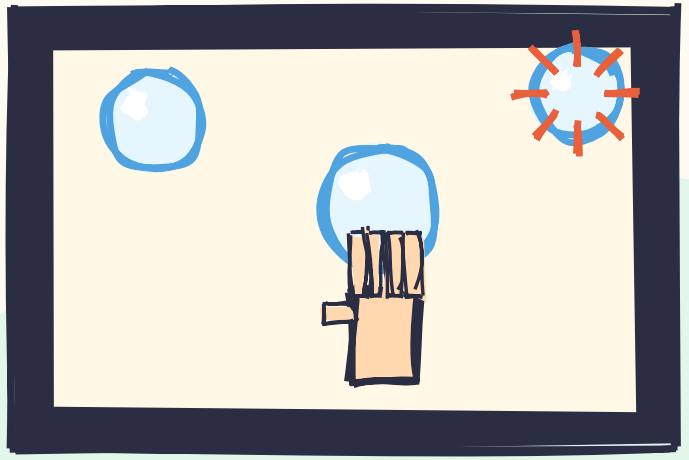
Little Explorers · Class 27

Pop the Bubbles with Your Hand

A game where YOU are the controller!

Camera Pop! Move & play



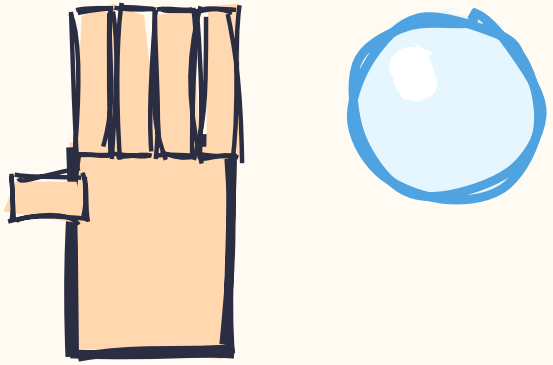
Did you know?
Some games use a camera to watch you jump and dance. Your body is the controller!

Let's wonder...

Can you play a game with no buttons?

Look, no hands... well, only hands!

Most games need buttons, a mouse or a tablet screen. What if the computer could **watch** you and you could play just by **waving**?



No way!

Need a mouse?

Wave at the camera!

Today: you build a **bubble-popping game** in Scratch that you play by moving your hands in front of the camera.

Learning objectives

Today I can...

1	**Camera sees:** say that the computer **watches movement** with the camera	**Proof:** I explained how my hand pops a bubble
2	**Pop!:** make bubbles **pop** when my hand moves over them	**Proof:** my bubbles pop when I wave
3	**Move and play:** play a game that gets my **body moving** and keep score	**Proof:** I popped 10 bubbles

Objective 3 · Move and play · video

Video time!



How Do Our Bodies Move?

SciShow Kids · about 4 minutes · youtu.be/j918PoWWaB0

Watch for...

1. What helps our body **move**?
2. What are **muscles** and **bones**?
3. How does your brain tell your arm to move?

Scan to watch

After the video: Wave your hand slowly, then fast. Today the computer will WATCH you move!

LEARN

Objective 1 · Camera sees

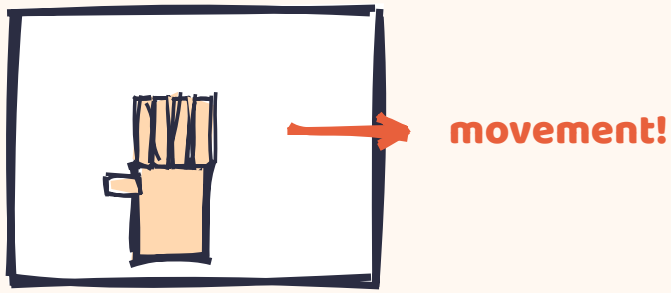
The camera watches movement

The idea

Scratch's **video sensing** looks at the camera picture many times a second. It notices where things **change**, like your hand moving.

If lots of movement happens on top of a bubble: **POP!**

Example: spotting a fly



Sit very still and look at a wall. If a fly lands, you notice it straight away, because something MOVED. The computer notices movement the same way.

Objective 2 · Pop!

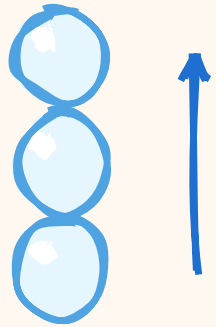
Bubbles that float

The idea

Make the bubble **float up** slowly: change y by 2, again and again. When it reaches the top, send it back to the bottom!

Now you must **chase** the bubbles. More moving = more fun!

Example: a real bubble



Blow a soap bubble: it floats up and away. You have to reach up to pop it. Our computer bubbles float too!

LEARN

Real world

Cameras that watch movement



Motion games
Dance and sports games watch you move



Exercise apps
Count your jumps and squats



Automatic doors
Open when they see you coming



Motion lights
Switch on when someone walks past

? Where have you seen something that reacts when you move?

Move more!
Moving games are good exercise. Your heart gets stronger every time you jump!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human bubble machine
2. Turn on video sensing
3. Snap the picture blocks
4. Make bubbles float and multiply
5. Add a timer
6. Score board

Class game

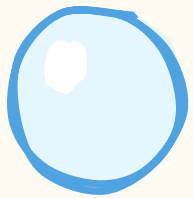
Human bubble machine

How to play

1. Half the class are **bubbles**: crouch with arms in a circle.
2. The other half are **poppers**.
3. A popper waves near a bubble: the bubble jumps up and shouts 'POP!' then sits down.
4. Swap after 1 minute!

Then discuss

What did a popper need to do to pop a bubble?



Big idea

The bubble only pops when it 'feels' movement right on top of it.

MAKE

Objective 1 · Build step 1 of 3

Turn on video sensing

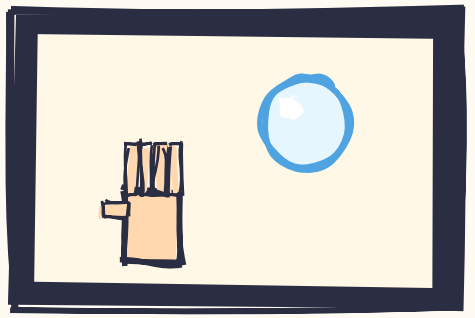
a Open **Scratch** on a computer with a camera.

b Click the **extensions** button (bottom left). Add **Video Sensing**.

c Allow the camera if it asks (teacher helps).

d You see yourself on the stage. Wave!

e Choose a **Ball** sprite and make it look like a bubble.



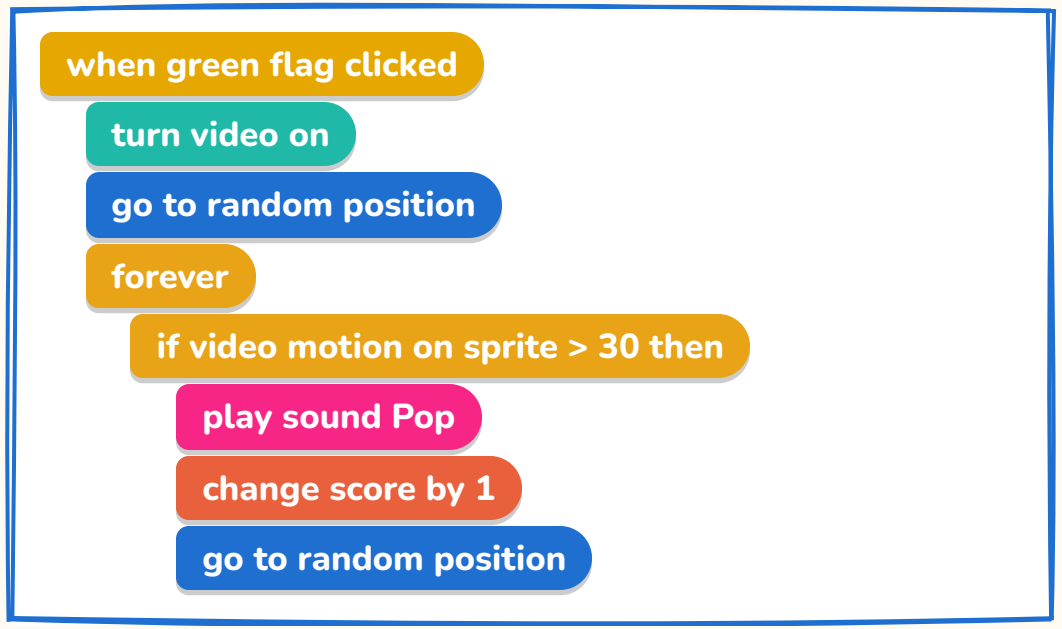
Check: You can see yourself on the Scratch stage.

Only use the camera in class with the teacher. Never record without asking.

MAKE

Time to build!

Snap the picture blocks



Read it like a story

Watch the camera.

If my hand moves on the bubble:

POP! Add 1 point.

Bubble jumps somewhere new!



Make a score

Click Variables → Make a Variable → call it 'score'.

MAKE

Objective 2 · Build step 2 of 3

Make bubbles float and multiply

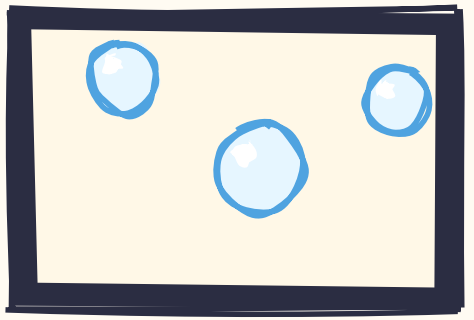
a Inside forever, add **change y by 2**: the bubble floats up.

b If y position > 170, **go to x: random, y: -170** (back to the bottom).

c Right-click the bubble: **duplicate** it 3 times.

d Make each bubble a different colour or size.

e Play: wave to pop!



Check: Lots of bubbles float up and pop when you wave.

MAKE

Objective 3 · Build step 3 of 3

Add a timer

a Make a variable called **time**. Set it to 30 at the start.

b Add: repeat until time = 0: wait 1 second, change time by -1.

c When time = 0: say 'Game over! Score: ' + score.

d Play! How many bubbles in 30 seconds?

e Swap players: who can beat the score?



Check: The game stops after 30 seconds and shows your score.

Play!

Score board

Player	Bubbles popped	Moves used
Player 1		
Player 2		
Player 3		

Talk about it

Did big arm moves or small hand moves pop more bubbles?

Light!
The camera works best in a bright room, with you facing the light.

Objective check

Can I really do it? Prove it.

01	How does the computer know your hand moved?	○ ○ ○
02	Show a bubble popping when you wave.	○ ○ ○
03	What was your best score in 30 seconds?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Bubble tournament: every child plays 30 seconds. Cheer for everyone!

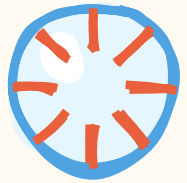
Show what you know

What did we learn?

- [x] Video sensing watches for movement
- [x] If movement is on the bubble: pop!
- [x] Variables keep the score and the time
- [x] Our bodies can be game controllers

At home

Play a game at home where you have to move, like Simon Says!



Next class: Cardboard Automaton!

Exit ticket: what does the camera look for?